

Blockchain and Its Role in the Decentralized Finance Sector

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Abstract

This report explores Blockchain and its impact on the Decentralized Finance (DeFi) Sector. The traditional finance systems rely heavily on the third party such as banks and brokerage, which is expensive, time-consuming, and inefficient. Now, with the rise of blockchain networks such as Bitcoin and Ethereum, the shift towards DeFi system has gained momentum.

DeFi uses blockchain based platforms to eliminate the third party by connecting everyone on the platform and reducing the cost while enhancing the transparency, accessibility, and efficiency. Through smart contracts and decentralized applications (DApps) users can lend, borrow, trade, and invest directly and securely.

This report talks about how DeFi works on smart contracts and what are the advantages as well as the risks, vulnerabilities by covering real life case studies and the regulation taken by government to reduce the risks. Despite these challenges, in my opinion DeFi is a powerful shift in how we approach finance and with further research and development the challenges can be overcome and help reshape the finance sector.

1 Introduction

In the last few years, blockchain has become one of the biggest innovations in the digital world. At first, it was just the technology behind cryptocurrencies like Bitcoin. But now, it's being used in many other areas. It brings more transparency, better security, and removes the need for middlemen. One area that's changing a lot because of the blockchain is the finance sector. This has led to the rise of Decentralized Finance (DeFi), where people can use financial services without banks or brokers.

The old system, called Centralized Finance (CeFi), depends on third parties like banks to handle transactions. While this system has worked for a long time, it also has problems. These include high fees, slow services, less access for people in remote areas, and lack of transparency. It also gives full control to one authority, which can be risky. That's why many people are now looking for a better and more open system.

That's where DeFi comes in. It runs on public blockchains like Ethereum and uses smart contracts to do things automatically. These smart contracts are like computer programs

that don't need a bank to process transactions. People can lend, borrow, trade, and invest directly using decentralized apps (DApps). Everything is open and recorded on the blockchain for everyone to see.

DeFi's main goal is to give people more control over their money, lower costs, and make finance easier for everyone. Unlike banks, DeFi works 24/7 and is open to anyone with internet and a digital wallet. I think this is very helpful for people who don't have access to banks in their area.

But DeFi also has risks. Since it's still new, there have been many security issues, bugs in the code, and even price attacks. Some platforms have been hacked, and there's also confusion about laws and regulations. So even though DeFi is growing, it still needs better safety and clear rules.

This report will explain how blockchain makes DeFi possible, how these platforms work, and how they are different from traditional finance. It will also look at real examples like the DAO Attack (2016) and Ronin Bridge Hack (2022) to show the challenges. At the end, we'll also look at what governments are doing about DeFi and what the future might look like.

To sum up, DeFi and blockchain are creating a new kind of finance that is open, global, and user-friendly. With stronger security and proper laws, I think DeFi could become a very important part of our future economy.

2 Objective

The main objective of this project is to study blockchain technology and how it is changing the world of Decentralized Finance (DeFi). It focuses on how blockchain removes middlemen like banks, which helps reduce costs, improve transparency, and make transactions more secure. This project also looks at the role of smart contracts and decentralized apps (DApps) in creating automated and trustless financial services. Another goal is to explain how DeFi is different from traditional finance and what benefits it offers. Along with the advantages, the project also talks about the risks, challenges, and legal issues that come with DeFi.

3 Literature Review

In September 2024, researchers Akhilesh A. Wao, Anshita Agrawal, and Harshvardhan Singh discussed the growing relevance of blockchain technology especially in how it's changing Decentralized Finance (DeFi). According to their study, DeFi is essentially a financial system built on public blockchains which means it doesn't rely on traditional intermediaries like banks or brokers. Instead, people can directly access financial services

through decentralized platforms. What makes DeFi stand out is its use of blockchain-based tools to turn traditional finance into more transparent and trustless systems. The aim of DeFi is to make financial services more open and accessible to everyone even those who are left out by conventional banking. One of the biggest advantages they highlighted is that cutting out intermediaries can make transactions faster and cheaper. Since everything is recorded on a public blockchain, DeFi systems are easier to track and audit than traditional banking methods and operate globally 24/7 without being limited by business hours. Smart contracts is an important key that automatically executes the agreements, making financial services more efficient and trustless. The paper also highlights the disadvantages such as scalability, regulations, security risks etc. Despite these challenges, in my opinion DeFi is a game-changing innovation. The paper concludes with the author mentioning that DeFi could lead to a more open, fair and resilient global financial system. [1]



(Figure 1: DeFi)

In 2020, researchers Dirk Zetsche, Douglas Arner, and Ross Buckley wrote about how DeFi is changing finance with the help of new technologies. According to the authors, DeFi lets people use financial services without needing intermediaries. It works using blockchain, smart contracts, and cloud systems. I think that DeFi is a smart step forward but it comes with risks. The authors say that even though it removes old control systems but it still needs new rules to keep things safe. In my opinion, the idea of putting regulation directly into the system code called embedded regulation is unique. The paper also says that DeFi could weaken legal control since it's hard to track who is in charge. I believe this could cause problems if no one is held responsible. The report also shows

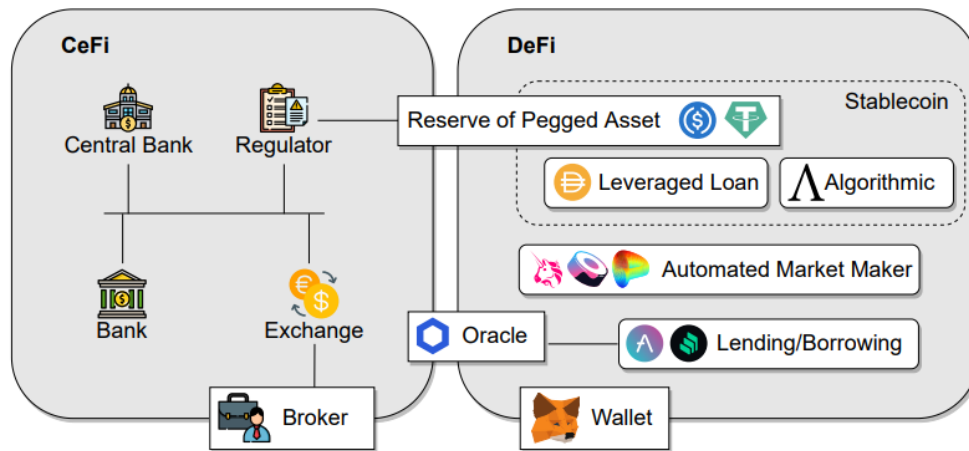
how DeFi could help people in areas where banking is not strong but it also warns about issues like data privacy, tech failures, and enforcement. I think if we combine innovation with smart regulation DeFi can be a revolutionary in a safe way. [2]

In 2021, Fabian Schär discussed how DeFi is changing finance by using smart contracts on blockchain platforms like Ethereum. The author says that DeFi doesn't need banks or intermediaries. It runs on code that anyone can check which is known as Smart Contract. I think that makes it more transparent and fairer. The report explains how smart contracts help create financial services like lending, exchanges, and investments without needing trust. In my opinion, this makes DeFi more open and fast compared to traditional finance. Schär also talks about five layers in DeFi settlement layer to aggregation layer. The paper also says tokenization helps turn real assets into digital tokens which makes trading quicker and easier. I found it interesting that DeFi apps are like Lego blocks people can combine different parts to build new systems. The author warns about risks too like bugs in smart contracts, admin control, and dependence on outside data also known as oracles. In my view, DeFi has great potential but needs better security and rules. The paper ends by saying if these issues are fixed DeFi could make global finance more open, strong and reliable. [3]

On July 31, 2023, authors Kose John, Leonid Kogan and Fahad Saleh explained how smart contracts are used in DeFi. This report says smart contracts are pieces of code that run on blockchains and help people make automatic and trust less transactions. In my opinion, they are useful because they remove the need for intermediaries. The authors say smart contracts help solve commitment problems and let users trade, lend, and borrow directly. They also explain that smart contracts face problems like not being able to get real-world data easily which is called the oracle problem. I think the paper shows how using smart contracts can lower costs but also introduces new risks like bugs, privacy issues, and storage costs. The paper also talks about stablecoins and tokens saying that not all are truly stable. In my view, smart contracts can improve finance, but only if we solve issues like security and legal limits. The authors end by saying DeFi can grow more if these challenges are managed properly. [4]

On 16 June 2021, researchers Kaihua Qin, Liyi Zhou and others studied the difference between Centralized Finance (CeFi) and Decentralized Finance (DeFi). This report explains that the Traditional system uses trusted third parties like banks while DeFi lets users control their own assets using smart contracts. In my opinion, DeFi is more user-friendly because people do not need permission to join or use it. The authors say that

DeFi is transparent and works all day unlike CeFi which runs on fixed hours. They also talk about stablecoins like USDT and USDC and say that these are often thought of as DeFi but they are actually controlled by companies. I think this is risky because these companies can freeze or erase funds. The paper shows a way to check whether something is really DeFi or still depends on a central authority. I think DeFi offers more freedom and higher returns but it also has problems like bugs and price attacks. The authors believe CeFi and DeFi should work together. In my view, combining both can help make the financial world safer and smarter. [5]



(Figure 2: Architecture of CeFi and DeFi)

In 2022, Peterson Ozili explored how decentralized finance is being researched and used across the world. The report mentions that DeFi is a system that runs on public blockchains and offers financial services without needing third parties. In my opinion, this gives more people a chance to access financial tools especially those who cannot use regular banks. The author explains that DeFi makes cross-border payments faster and cheaper, encourages innovation and improves transparency. He also talks about some big risks like coding mistakes in smart contracts, legal confusion, cyber threats, and the misuse of DeFi for illegal purposes. I think this shows that while DeFi has many advantages it also needs strong safety steps. According to the paper, DeFi is becoming popular in regions like the US, Asia and Europe but faces slow growth in Africa due to tech gaps and strict regulations. In my view, with better understanding and stronger policies DeFi can improve financial systems in many countries. [6]

In 2021, Patrick Schueffel introduced DeFi as a growing innovation that challenges the traditional banking system. This report says that DeFi allows people to access financial tools without going through banks, brokers, or insurance companies which makes it easier for people to manage their own money. The author explains that DeFi works on blockchain and uses smart contracts which automatically complete transactions based on code. What makes DeFi special is that it's not only automated but it is also decentralized meaning no single group controls it. This removes the chance of failure at one point and reduces trust issues. I found it interesting that DeFi is anonymous users and do not have to give their identity. The report also highlights that DeFi is permissionless and open source so anyone can build apps or services on it. Another unique point is that DeFi doesn't care about the product being traded or who the user is, what matters is that the process runs smoothly. In my view, DeFi is also more secure since data is stored on many nodes and can't be easily changed. However, the paper warns that DeFi is still unregulated which can be risky. Still, I believe it opens doors for people around the world who have never had access to finance before. [7]

In 2023, Bodo and Filippi discussed how trust in DeFi and blockchain is shaped by regulation. The paper explains that blockchain was created to remove the need for centralized trust and replacing it with code. Although in reality, people still rely on platforms, developers and external data sources. The paper highlights that the regulation can either build or break trust depending on how it's applied. If it is too strict, it can drive people away from DeFi. If it is too loose, it risks scams and fraud. The authors say contextual trust is key. People trust different parts of the system based on their role like users, developers, or investors. One new point is how smart contracts don't always remove the need for human judgment. The paper also talks about how DeFi must balance being trustless while still needing governance. In my view, this shows that regulation should support trust and not control everything. [8]

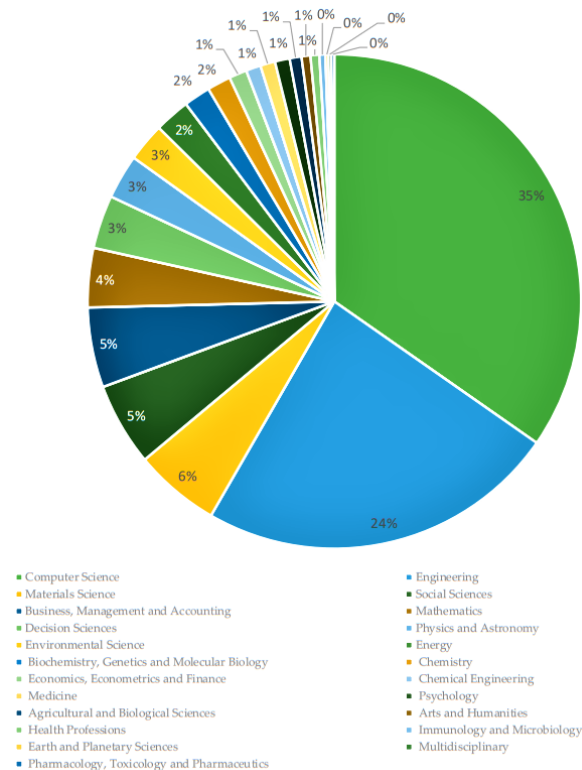
In 2023, Hamed Taherdoost mentions in his report that smart contracts are like digital agreements that run on their own using blockchain. This removes the need for banks, agents and lawyers. The report says they are safe because no one can change them once they are created. I think this makes the system more secure but also risky if there's a coding mistake. The author talks about how smart contracts are used in many fields like finance, education, healthcare, and voting.

The pie chart in the report clearly shows how smart contracts are divided across different sectors. 35% of smart contracts are used in computer-related work, which is more than any other field. This includes areas like software development, cybersecurity, cloud

computing, and automation tools. I think this proves that computer science and engineering is the leading field when it comes to adopting blockchain.

The chart also shows that 25% are used in education, such as in certificate issuing, managing student records, or online exam systems. Then comes healthcare with 15%, which could be for storing patient data or hospital records. Financial services hold 10%, including online banking, crypto loans, and trading platforms. Voting systems make up 5%, which means governments or organizations are starting to evaluate smart contracts for safe and fair elections. The last 10% is marked as “others”, which may include law, real estate, and logistics.

In my view, this chart makes it easy to understand where smart contracts are really growing. The fact that the largest portion is in computers shows how much developers and tech industries trust this system. The report also says bugs and lack of rules are still big problems. So, I believe that smart contracts should always be evaluated and reviewed properly. If used correctly, they can make systems faster, cheaper, more secure, and more reliable. [9]



(Figure 3: Sector-Wise Distribution of Smart Contract Usage)

Muhammad Izhar Mehar, Charles Shier, Alana Giambattista, and their team discussed the famous DAO Attack and its effect on Ethereum in their 2021 paper. The DAO was introduced in 2016 as a decentralized system for funding projects using Ethereum's blockchain. It allowed users to invest Ether and vote on which projects should receive funds through smart contracts. I think this was a creative idea but the system had a major flaw in its code. On June 17, 2016, a hacker found a loophole that let them withdraw funds multiple times before the balance updated. Over \$50 million USD worth of Ether was taken and stored in a child DAO for 28 days. The Ethereum community had to make a big decision: ignore the attack, freeze the stolen funds (soft fork) or reverse the chain (hard fork). They finally voted for a hard fork which returned the stolen Ether to the investors. However, some people disagreed and continued using the old chain, now called Ethereum Classic. In my opinion, this was a key moment in blockchain history. It showed the dangers of smart contract bugs and made people question whether code should always be treated as law. This incident also proved that mistakes in code can teach us valuable lessons and remind us to be more careful in the future. [10]



(Figure 4: Timeline of Dao Attack)

In 2023, Nikita Belenkov and team studied many bridge hacks, including the Ronin Bridge attack from 2022. Ronin was a bridge used by the game Axie Infinity to transfer ETH and USDC between Ethereum and the Ronin network. In March 2022, a hacker stole 173,600 ETH and 25.5 million USDC which was worth over \$600 million USD. The attack happened because the hacker got access to 5 out of 9 validator keys which were needed to approve withdrawals. In my opinion, this shows the risk of centralization. The attacker used a backdoor through a gas-free RPC node that was still active from a past permission setup. Using those keys, they made fake transactions and stole the funds. What's worse is the hack wasn't discovered until 6 days later. I think this delay made things worse. This case proved that smart contracts weren't the problem, private key

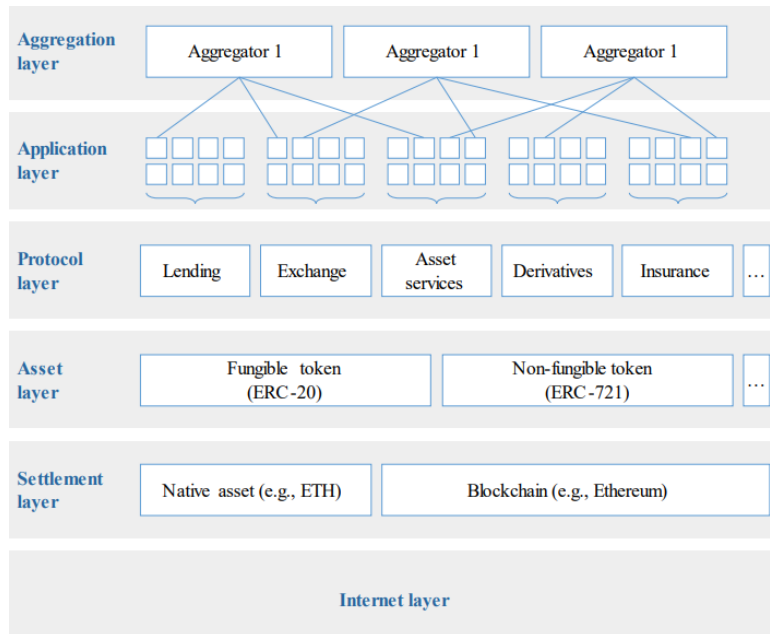
security was. In my view, the Ronin hack showed how dangerous weak validator setups can be. Decentralizing validators and better key management could have stopped this. [11]

In July 2022, CMA (Dr.) Paritosh Basu wrote about how DeFi is changing the world of finance. The article compares Centralised Finance (CeFi) with Decentralised Finance (DeFi) and explains how DeFi grew after the 2008 financial crisis. In my opinion, the author shows how people lost trust in banks and looked for something new. DeFi removes intermediaries and uses smart contracts to carry out transactions. The article explains how DeFi is open, permissionless and based on blockchain. It also highlights that DeFi users control their own assets and don't need to trust a central authority. The author says DeFi has features like encryption, privacy and peer-to-peer transactions. The paper also talks about CeFi like services inside DeFi platforms like crypto loans and insurance. It warns that DeFi has risks like market crashes, weak regulations and smart contract flaws. In my view, the article perfectly explains how DeFi works and why strong global rules are now needed. [12]

On 28th April 2025, Rula Almadadha mentions that blockchain helps banks improve their services by reducing costs, improving speed and cutting down errors. I think this is important because it shows how traditional banking can also benefit from new technology. The study uses real financial data and performance indicators like ROA (Return on Assets) and ROE (Return on Equity). The results showed a positive link between blockchain adoption and better performance. What I found new is that the report also highlights how blockchain builds stronger trust with customers by offering transparency and reducing fraud. The paper also says blockchain helps in decision-making by giving banks real-time and accurate data. However, it also talks about problems like high setup costs, unclear rules and the risk of cyberattacks. Another interesting point is that banks using blockchain were able to offer more digital products and compete better in the market. I believe that if these challenges are solved blockchain can be a major tool for growth in the banking industry. [13]

In 2022, Axel Wieandt and Laurenz Heppding explored the differences between Centralized Finance and Decentralized Finance. The paper starts by explaining how the 2008 financial crisis exposed the problems in CeFi especially its overdependence on large banks. In my opinion, this is where DeFi comes in as a solution. DeFi removes intermediaries and uses smart contracts to offer services like lending, borrowing and trading. The report says DeFi is faster, more transparent, and more accessible than CeFi.

They describe the DeFi Layer Model including blockchain infrastructure, tokens, smart contracts, and apps. Popular examples like MakerDAO, Aave, Uniswap, and Curve are discussed. In the authors' view, both DeFi and CeFi have pros and cons. CeFi is still important because it handles monetary policy but DeFi offers freedom and decentralization. The authors suggest that instead of replacing CeFi, DeFi may merge with it. In my view, this paper shows that the future of finance could be a mix of both systems. They also say risks like scalability, security flaws, and regulations need more attention. [14]

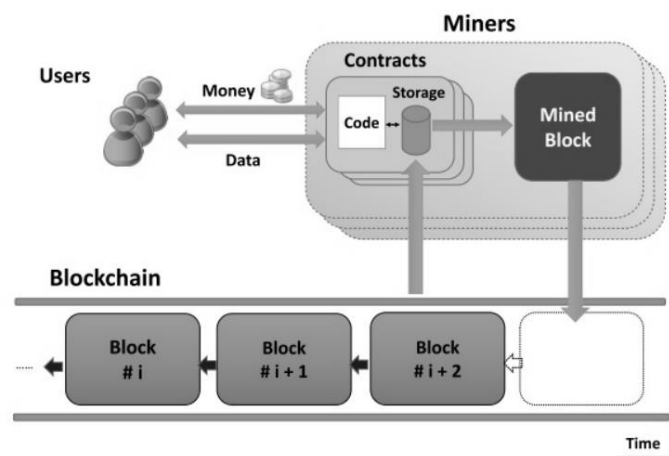


(Figure 5: DeFi layer model)

In 2024, Shubhi Rathore and Anandini Gupta shared some new views on DeFi that other papers didn't cover. They explained how DeFi is entering the gaming industry where players earn crypto instead of points. The paper talks about DeFi insurance where people get coverage through shared pools not big companies. They also highlight that user experience is still difficult and many platforms are not beginner friendly. One unique point is the mention of rural awareness showing how DeFi hasn't reached small towns yet. The report includes a survey that shows only 26% of people understand DeFi even though 74% know about crypto. They also found that people still trust banks more, even if they face issues like failed transactions. Another rare point is that DeFi could help people who face problems with debit and credit card payments. The paper suggests that DeFi needs a better interface and education to reach the common public. I think this makes it clear that DeFi's growth depends not just on technology but also on how easy it is for people to use. [15]

In 2024, Shashank Singh wrote about how DeFi is changing finance with the help of blockchain and cryptocurrency. The paper explains that smart contracts make services like lending, trading, and saving automatic and trustless. One unique point is that the author talks about Decentralized Identity and insurance protocols which are not covered in most reports. The paper also highlights the move toward eco-friendly systems like proof-of-stake (POS) to make DeFi greener. Another interesting point that the author mentions is of cross-chain connections which help different blockchains work together. The author warns about risks like code bugs and legal issues but also says that global rules are slowly developing. In my opinion, the paper shows that DeFi is more than just a tech trend but it's a step toward a fairer and more open financial world. [16]

In 2017, M. Alharby and A. van Moorsel conducted a detailed study on blockchain-based smart contracts. The paper maps 24 research papers shows that most of them focus on smart contract issues. Four main technical problems are highlighted: codifying, security, privacy, and performance. The authors explain that writing correct smart contracts is difficult and fixing bugs after deployment is almost impossible. One new point I found was how tools like OYENTE, HAWK and Town Crier are helping improve contract security, privacy and access to data. The paper also talks about smart contract applications like IoT, music rights, and identity systems. I think the study is useful because it shows gaps in research, especially on criminal activities, parallel execution and exploring blockchains other than Ethereum. In my opinion, future studies should focus more on performance issues and real-world applications to make DeFi safer and scalable. The paper ends by urging researchers to improve tools, language design and security testing to make smart contracts more reliable. [17]



(Figure 6: Smart Contract system)

In this report, Nic Carter explains how DeFi is different from both traditional finance and open banking. While open banking connects banks through APIs DeFi uses smart contracts on public blockchains, allowing people to access financial services without needing permission or identity verification. I found it interesting that DeFi lets users switch platforms easily since all apps run on shared blockchain code. The report says this openness gives power back to users but it also invites risk and makes it hard to regulate. DeFi apps act like Lego blocks meaning anyone can build on existing code without asking. I also learned that DeFi allows automatic trading through Decentralized exchanges where you don't have to give your money to a middleman. In my opinion, this makes DeFi faster and more secure. Another point the author made is about composability where one DeFi app can depend on others and create powerful financial tools but this also means if one part fails, it can affect the whole system. Still, I think the idea of DeFi giving people more control over their money is a good step. The author believes that as DeFi grows it could mix with traditional finance but this may remove some of its key benefits like full freedom and privacy. [18]

In 2024, Harshil Sharma and Shipra Agarwal explored how DeFi is transforming traditional financial systems. The paper says DeFi enables services like lending, trading, and asset management without banks, using smart contracts and blockchain. The study shows that DeFi supports real-time peer-to-peer lending and borrowing even for large amounts. The study highlights benefit like decentralization, transparency, and financial inclusion especially in developing countries. I found it interesting that the authors link DeFi with broader fintech trends like AI, big data and RegTech. The paper also warns about common risks such as smart contract bugs, security breaches and governance issues. They mention that many DeFi apps struggle with liquidity and scalability. Another new point is how they suggest regulatory sandboxes and standardized frameworks to balance innovation and safety. Real-world case studies on stablecoins, yield farming and DEXs are also included to show impact. The authors believe DeFi could change finance permanently if paired with clear regulation and better tech. [19]

In This report explains how blockchain is changing digital businesses, especially through DeFi. It talks about how traditional businesses, which depend on centralized control and middlemen are slowly being replaced by blockchain-based systems that are more secure, fast and transparent. The authors describe how blockchain allows people to do direct transactions without needing banks, lawyers or brokers. DeFi is one of the key examples, letting users borrow, lend, or invest using smart contracts. New terms like BusiFi (business finance), SciFi (science finance) etc. show how different sectors are adopting blockchain for better efficiency. Another interesting aspect is how DAOs can help create

a more democratic decision making process in companies. The authors argue that DeFi isn't just for big investors but it can also help normal people access better services. Although the paper also discusses the Metaverse, its most valuable insights come from how blockchain and DeFi can reshape work, payments and business trust. In my view, this shows that blockchain is not just a tool but it's becoming the backbone of future business. [20]

3.1 Summary of Literature Review

Reference No.	Author's Name	Year	Dataset	Analysis
[1]	Akhilesh A. Wao, Anshita Agrawal, Harshvardhan Singh	September, 2024	NA	DeFi uses blockchain and smart contracts to offer open, fast, and transparent financial services without banks. The study highlights both its benefits and risks, showing DeFi's potential to change global finance.
[2]	Dirk A. Zetzsche, Douglas W. Arner, and Ross P. Buckley	September 30, 2020	NA	The paper shows that while DeFi removes intermediaries but it still needs strong rules to stay safe. The idea of embedded regulation putting laws into code is a unique approach but also risky. Overall, the authors believe DeFi can grow if innovation and regulation are balanced properly.
[3]	Fabian Schär	February 4, 2021	NA	Fabian Schär highlights how DeFi's layered structure and tokenization make finance faster and more flexible. The Lego block approach shows how DeFi systems are customizable and open source. However, the paper also warns that without fixing risks like bugs and oracle issues DeFi's growth could be limited.
[4]	Kose John, Leonid Kogan, and Fahad Saleh	July 31, 2023	NA	The paper explains that smart contracts make DeFi more efficient by removing intermediaries and solving trust issues. However, challenges like

				the oracle problem, bugs, and unstable tokens still limit their full potential. The authors suggest that DeFi can only grow safely if these technical and legal risks are effectively managed.
[5]	Kaihua Qin, Liyi Zhou, Yaroslav Afonin, Ludovico Lazzaretti, Arthur Gervais	June 16, 2021	Figure 2: Architecture of CeFi and DeFi	The paper clearly shows that while DeFi gives users more control and freedom. Some parts like stablecoins still rely on central authorities. The authors introduce a method to test if a platform is truly decentralized which is very useful. They suggest that combining CeFi's safety with DeFi's innovation could create a better financial system.
[6]	Peterson K. Ozili	August 22, 2022	NA	Ozili highlights that DeFi has the power to improve global finance especially in underbanked regions. However, the lack of real-world research and the rise of cyber threats show that DeFi still faces major challenges. The paper also points out DeFi's uneven global growth, stressing the need for better education and stronger policies.
[7]	Patrick Schueffel	2021	NA	Patrick Schueffel shows that DeFi's biggest strength is its decentralization and permissionless nature, allowing anyone to use or build on it without needing approval. The paper highlights anonymity and the idea that DeFi focuses on how things work and not who is using it. While unregulated systems bring risk, they also offer financial access to people previously left out.
[8]	Balazs Bodo, Primavera de	2023	NA	Bodó and de Filippi says that DeFi still depends on human

	Filippi			trust despite its trustless nature. They show that regulation should focus on supporting trust not replacing it with strict control.
[9]	Hamed Taherdoost	February 13, 2023	Figure 3: Sector-Wise Distribution of Smart Contract Usage (Pie chart)	The report shows that computer science leads in smart contract usage, making up 35%, which proves how trusted and widely adopted it is in tech fields. The detailed pie chart helps visualize how smart contracts are growing across multiple industries. However, the report also warns that without strong rules and testing, smart contracts can still carry major risks.
[10]	Muhammad Izhar Mehar, Charles Shier, Alana Giambattista, Elgar Gong, Gabrielle Fletcher, Ryan Sanayhie, Henry M. Kim, Marek Laskowski	2021	Figure 4: Timeline of Dao Attack (Case Study)	The DAO Attack revealed how a simple smart contract bug could cause massive financial loss, even in a decentralized system. The community's decision to fork Ethereum raised ethical questions about whether "code is law" should always apply. This event became a turning point, showing the urgent need for better security and governance in DeFi
[11]	Nikita Belenkov, Valerian Callens, Alexandr Murashkin, Kacper Bak, Martin Derka, Jan Gorzny, Sung-Shine Lee	January 6, 2025	Case study	Nikita Belenkov and team analyzed the Ronin Bridge hack, where over \$600 million was stolen due to poor validator key management. The case highlights that the real issue wasn't smart contracts but centralized control and weak security.
[12]	CMA (Dr.) Paritosh Basu	July, 2022	NA	The author explains how DeFi offers more control and privacy by removing banks and using smart contracts. The article shows how DeFi grew after the 2008 crisis and calls for strong global regulations to handle risks like crashes and code

				flaws.
[13]	Rula Almadadha	April 28, 2025	NA	Rula's study shows that blockchain not only boosts bank performance but also improves customer trust and decision-making. It proves that even traditional banks can benefit from adopting modern technology. However, high costs and security risks still need to be addressed for long-term growth.
[14]	Axel Wieandt, Laurenz Heppding	2022	Figure 5: DeFi layer model	Authors compare CeFi and DeFi showing how DeFi offers more freedom, speed, and transparency. They believe the future of finance may be a hybrid system combining the strengths of both while fixing risks like scalability and regulation.
[15]	Shubhi Rathore and Anandini Gupta	January, 2024	NA	Shubhi Rathore and Anandini Gupta highlight unique points like DeFi in gaming, DeFi insurance and low rural awareness. Their survey shows that lack of understanding and poor user experience are big barriers to DeFi's growth.
[16]	Shashank Singh	January, 2024	NA	It explains how DeFi is improving financial services using blockchain and smart contracts. The paper shares new ideas like decentralized identity, insurance protocols, and eco-friendly blockchain systems. It shows that DeFi is growing fast but still needs better rules and security.
[17]	Maher Alharby, Aad van Moorsel	2017	Figure 6: Smart Contract system	Authors analyzed 24 papers and found that most smart contract research focuses on technical issues like bugs, security flaws and performance limits. They suggest better tools and more research into criminal activity, other blockchain platforms and

				real-world contract use.
[18]	Nic Carter	2020	NA	Researcher discusses how DeFi gives users financial freedom by removing middlemen and letting apps work together like building blocks. The report shows both the promise and risks of this system, especially around transparency, security and regulation.
[19]	Harshil Sharma, Shipra Agarwal	2024	NA	Sharma and Agarwal explain how DeFi is reshaping traditional finance using smart contracts and blockchain. They highlight benefits like inclusion and automation but also warn about risks like bugs, lack of regulation, and liquidity issues. Regulatory sandboxes and global frameworks are suggested as solutions.
[20]	Saeed Banaeian Far, Azadeh Imani Rad, Maryam Rajabzadeh Asaar	2023	NA	This paper highlights how blockchain and DeFi can replace centralized financial systems, empowering small businesses and individuals by removing middlemen. It also introduces new subfields like BusiFi and RegFi that could transform how companies operate and collaborate.

4 Applications

Blockchain technology supports various DeFi applications across multiple sectors. The most common use is in lending and borrowing where users can deposit crypto as collateral and get loans instantly through smart contracts. Another major application is decentralized exchanges (DEXs) like Uniswap, Aave etc. which allow users to trade tokens without intermediaries. DeFi also enables stablecoins, digital currencies tied to real-world assets like the US dollar used for payments and savings.

In the insurance sector DeFi protocols help users create peer-to-peer coverage models without companies. Gaming and NFT marketplaces use DeFi tools for in-game purchases and asset trading. Another growing area is cross-border payments which are faster and

cheaper than traditional banking systems. Some projects also explore decentralized identity systems and DAO-based governance where users vote on changes in the platform.

These applications show how DeFi is not just about finance but can improve many areas of digital life. The flexibility, low cost, and speed of these services make DeFi a strong alternative to traditional systems.

5 Issues and Challenges

Despite its advantages, DeFi faces several issues and challenges that need to be addressed. One major concern is the security risk in smart contracts. If the code has bugs it can be exploited by hackers as seen in cases like The DAO attack and Ronin Bridge hack.

Another big issue is the oracle problem where smart contracts rely on external data sources that may be inaccurate or manipulated. Lack of regulation is also a concern since DeFi platforms operate globally, there's no clear legal authority to monitor or protect users. User awareness and technical knowledge are low, making DeFi platforms hard to use for ordinary people. Price volatility in crypto tokens can lead to sudden losses, which reduces trust in the system. Liquidity problems in smaller platforms also make large transactions difficult.

Additionally, cross-chain risks arise when DeFi operates across multiple blockchains without proper coordination. All these challenges show that although DeFi is promising, it is still in a developing stage and needs better safety measures, education, and governance to reach mainstream adoption.

6 Conclusion and Future Scope

6.1 Conclusion

Blockchain and decentralized finance (DeFi) are creating a major shift in how the world thinks about money and financial systems. By removing intermediaries and using smart contracts, DeFi gives people more control, lower costs, and better access to services like lending, trading, and insurance. It also promotes transparency and security by recording all transactions on public blockchains.

However, DeFi still has challenges such as smart contract bugs, legal gaps, and lack of user education. Real-life cases like the DAO and Ronin hacks show how important strong coding and regulation are. Even so, DeFi continues to grow rapidly, with more users and

investors joining every year. It is also finding new applications in areas like gaming, insurance and governance.

In conclusion, DeFi has the potential to create a more open, fair, and global financial system. With ongoing improvements in security, technology, and regulation, blockchain-powered DeFi can become a trusted and essential part of the future economy.

6.2 Future Scope

The growth of blockchain technology shows strong potential to further change how financial systems operate especially through Decentralized Finance. While DeFi has already introduced a new way of managing and accessing financial services without banks or intermediaries, its future promises even more advanced and inclusive systems.

In the coming years, DeFi platforms are expected to improve in terms of safety, design and ease of use. Developers are working on making smart contracts more reliable so that the risks of bugs and attacks are minimized. One area gaining attention is cross-chain compatibility which would allow different blockchain networks to work together. This can help expand DeFi's reach and make the system more flexible and powerful.

As technology becomes more popular, the need for proper regulation is increasing. While DeFi is built to reduce third-party control, some basic rules may help protect users and reduce fraud without affecting decentralization. A balanced legal framework could help more people feel confident using DeFi.

In the future, DeFi may also expand into areas beyond finance. It could be used in identity verification, supply chain tracking, insurance and digital ownership. The use of Decentralized Autonomous Organizations or simply known as DAOs could help people participate in decision-making across platforms. There is also a growing focus on using energy-efficient blockchain systems which is important for long-term sustainability.

Overall, DeFi is still evolving but with continuous improvements in technology and responsible development, it could shape a more open, fair and global financial system in the future.

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8 Availability Statement

No new datasets were generated or analyzed in this chapter. All sources are cited and publicly accessible via the references listed.

9 References

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10 List of Illustrations

Figure No.	Title	Source
Figure 1	DeFi	https://forkast.news/wp-content/uploads/2022/11/Forkast-Opinion-Images-14-1260x710.png
Figure 2	Architecture of CeFi and DeFi	K. Qin, L. Zhou, Y. Afonin, L. Lazzaretti, A. Gervais (2021). <i>CeFi vs. DeFi – Comparing Centralized to Decentralized Finance</i> . https://arxiv.org/abs/2106.08157
Figure 3	Sector-Wise Distribution of Smart Contract Usage	H. Taherdoost (2023)- <i>Smart Contracts in Blockchain Technology: A Critical Review</i> . Information 2023, 14, 117. https://doi.org/10.3390/info14020117
Figure 4	Timeline of Dao Attack	M. I. Mehar, C. Shier, A. Giambattista, E. Gong, G. Fletcher, R. Sanayhie, H. M. Kim, M. Laskowski (2021). <i>Understanding a Revolutionary Blockchain Attack – The DAO Hack</i> . SSRN Electronic Journal. https://doi.org/10.2139/ssrn.3164170
Figure 5	DeFi layer model	Wieandt, A., & Heppding, L. (2022). <i>Centralized and Decentralized Finance: Coexistence or Convergence?</i> SSRN. https://doi.org/10.2139/ssrn.4140820
Figure 6	Smart Contract system	M. Alharby and A. van Moorsel, Blockchain-Based Smart Contracts: A Systematic Mapping Study, published in <i>Computer Science & Information Technology (CS & IT)</i> , Vol. 7, No. 10, 2017.